Track 1: Artificial Intelligence and Machine Learning (AI/ML)

This track invites participants to apply data-driven intelligence to decision-making in agriculture. Solutions should be practical for smallholder and commercial farmers, should explain their outputs in plain language, and should acknowledge the uncertainty that is inherent in any prediction.

Problem Title 1: AI-Powered Farmer Assistant

A. Problem Description

Farmers routinely make decisions about sowing time, irrigation, fertilizer doses, and pest control with limited access to timely expert advice. Extension officers serve large numbers of villages, and printed or online advisories are often generic, written in a language the farmer does not use daily, or unavailable when a question arises in the field.

As a result, a farmer growing paddy or coconut may depend on neighbours, input dealers, or past habit, which can lead to avoidable crop losses and excess input costs. A digital assistant that answers questions in a familiar language and tailors guidance to the crop and local context could narrow this gap.

B. Challenges

1. Obtaining reliable, verified agricultural knowledge for the assistant and keeping it current across crops, seasons, and regions.
2. Supporting regional languages, dialects, and colloquial farming terms in text and, where feasible, voice input.
3. Preventing incorrect or unsafe advice, such as wrong pesticide doses, through grounding, validation, and clear limits on what the assistant answers.
4. Operating in areas with low bandwidth or intermittent connectivity and on low-cost smartphones.
5. Designing an interface suitable for users with limited digital or functional literacy.

Problem Title 2: AI-Based Crop Disease and Pest Prediction

A. Problem Description

Crop diseases and pests often spread quickly, and by the time symptoms are obvious the damage may already be widespread. Diseases such as blast in paddy or leaf spots in vegetables also depend on weather conditions such as humidity and temperature, which can signal risk before visible symptoms appear.

Farmers frequently find it difficult to tell similar-looking symptoms apart, and laboratory diagnosis is slow or distant. An early-warning tool combining leaf images with weather and field data could help farmers decide when to inspect and act.

B. Challenges

1. Collecting labelled image datasets that reflect real field conditions rather than laboratory backgrounds.
2. Handling variation in lighting, camera quality, growth stage, and overlapping symptoms or nutrient deficiencies that resemble disease.
3. Combining image-based findings with weather and historical outbreak data in a meaningful way.
4. Communicating confidence levels honestly and avoiding overconfident or false alerts that erode trust.
5. Delivering results on low-end devices and with limited connectivity.

Problem Title 3: AI-Powered Farmer-to-Market Connectivity

A. Problem Description

Many farmers sell through a few local intermediaries and have limited knowledge of prices in other markets or of buyers who may offer better terms. Prices vary by location, quality, and season, and perishable

produce must be sold quickly, which weakens the farmer's bargaining position.

A platform that gathers price information, matches produce with interested buyers, and compares options after accounting for transport and timing could support more informed selling decisions.

B. Challenges

1. Sourcing reliable, current price data from markets and public portals, which may be incomplete or inconsistently formatted.
2. Accounting for quality grades, transport cost, perishability, and quantity when comparing markets.
3. Building trust between farmers and buyers, including verification of buyers and clarity on payment terms.
4. Making the platform usable for farmers with limited literacy and in regional languages.
5. Remaining affordable and workable where internet access is intermittent.

Problem Title 4: AI-Based Crop Yield Prediction

A. Problem Description

Yield varies widely with season, variety, soil, and management. Without a reasonable estimate in advance, farmers and local aggregators find it hard to arrange labour, storage space, transport, and buyers, and surpluses or shortfalls cause avoidable losses.

A data-driven estimate, even an approximate one with a stated range, can support planning for harvest and marketing.

B. Challenges

1. Limited availability of reliable, farm-level historical yield records, especially for smallholdings.
2. Combining heterogeneous data, such as weather, soil, and management practices, collected at different scales and frequencies.

3. Avoiding overfitting when data is scarce and ensuring the model generalises across regions and seasons.
4. Accounting for unpredictable events such as extreme weather or disease outbreaks.
5. Presenting predictions as ranges in a form that farmers and planners can use.

Problem Title 5: AI-Based Livestock Disease Prediction and Early Warning

A. Problem Description

Livestock diseases can cause significant economic losses for farmers by reducing milk production, affecting animal health, and increasing veterinary expenses. In many rural areas, farmers may not recognize early symptoms of diseases, resulting in delayed treatment and the spread of infections among animals.

Develop an AI/ML-based system that helps farmers identify potential livestock health problems at an early stage. The system should analyze available information, such as animal health records, symptoms, body temperature, feeding patterns, activity levels, and milk production changes, to identify possible disease risks. It should provide early warnings and recommendations to help farmers seek timely veterinary assistance.

B. Challenges

1. Identifying meaningful patterns in livestock health and behavioural data.
2. Predicting potential disease risks using incomplete or limited datasets.
3. Distinguishing normal variations in animal behaviour from possible disease symptoms.
4. Providing reliable early warnings while minimizing false alarms.
5. Designing a simple, user-friendly system for farmers with limited technical knowledge.

6. Supporting different livestock types and varying farm conditions.

Problem Title 6: AI-Powered Weather and Climate Risk Prediction

A. Problem Description

Weather extremes such as prolonged dry spells, intense rainfall, or heat waves can damage crops at sensitive growth stages. General forecasts rarely indicate what a given condition means for a particular crop or stage, so farmers may not know what action to take or when.

A system that translates weather data into crop-specific risk indications and preparedness suggestions can help farmers protect crops and plan field operations.

B. Challenges

1. Obtaining forecasts and historical data at a resolution that is relevant to individual farms.
2. Defining crop- and stage-specific risk thresholds that are scientifically sound.
3. Managing forecast uncertainty and avoiding frequent false alarms that cause alert fatigue.
4. Delivering alerts reliably through channels farmers use, such as SMS or voice messages.
5. Providing recommendations that are practical and affordable for small farmers.